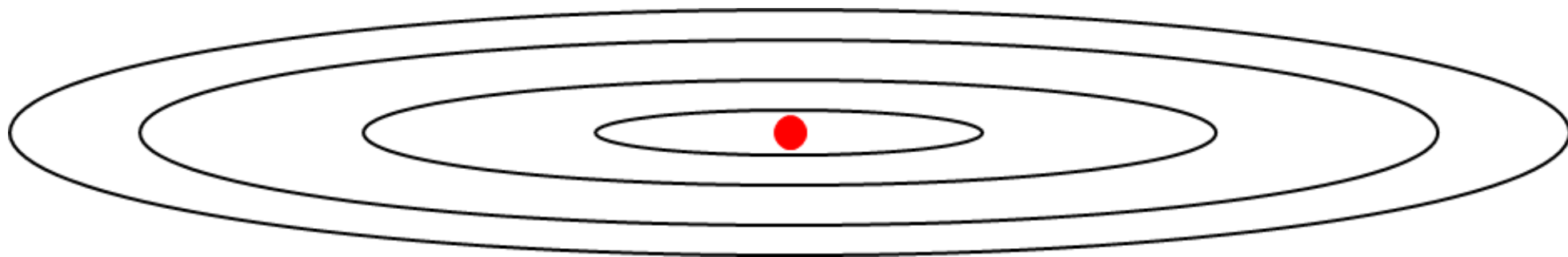


Momentum



On iteration t :

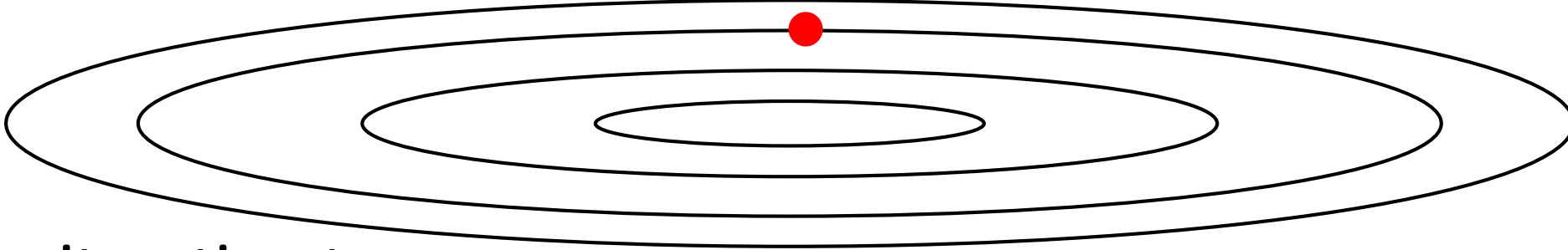
Compute dW, db on the current mini-batch

$$v_{dW} = \beta v_{dW} + (1 - \beta) dW$$

$$v_{db} = \beta v_{db} + (1 - \beta) db$$

$$W = W - \alpha v_{dW}, \quad b = b - \alpha v_{db}$$

RMSprop: Root Mean Square Prop



On iteration t

Compute dW and db on the mini batch

$$s_{dW} = \beta_2 s_{dW} + (1 - \beta_2) dW^2$$

$$s_{db} = \beta_2 s_{db} + (1 - \beta_2) db^2$$

$$W = W - \alpha \frac{dW}{\sqrt{s_{dW} + \epsilon}}$$

$$b = b - \alpha \frac{db}{\sqrt{s_{db} + \epsilon}}$$

Adam optimization algorithm

- Hybrid of Momentum and RMSProp

$$s_{dW} = 0, v_{dW} = 0, s_{db} = 0, v_{db} = 0$$

On iteration t

Compute dW and db on the mini batch

$$s_{dW} = \beta_2 s_{dW} + (1 - \beta_2) dW^2, v_{dW} = \beta_1 v_{dW} + (1 - \beta_1) dW$$

$$s_{db} = \beta_2 s_{db} + (1 - \beta_2) db^2, v_{db} = \beta_1 v_{db} + (1 - \beta_1) db$$

$$s_{dW}^{corrr} = \frac{s_{dW}}{1 - \beta_2^t}, s_{db}^{corrr} = \frac{s_{db}}{1 - \beta_2^t}$$

$$v_{dW}^{corrr} = \frac{v_{dW}}{1 - \beta_1^t}, v_{db}^{corrr} = \frac{v_{db}}{1 - \beta_1^t}$$

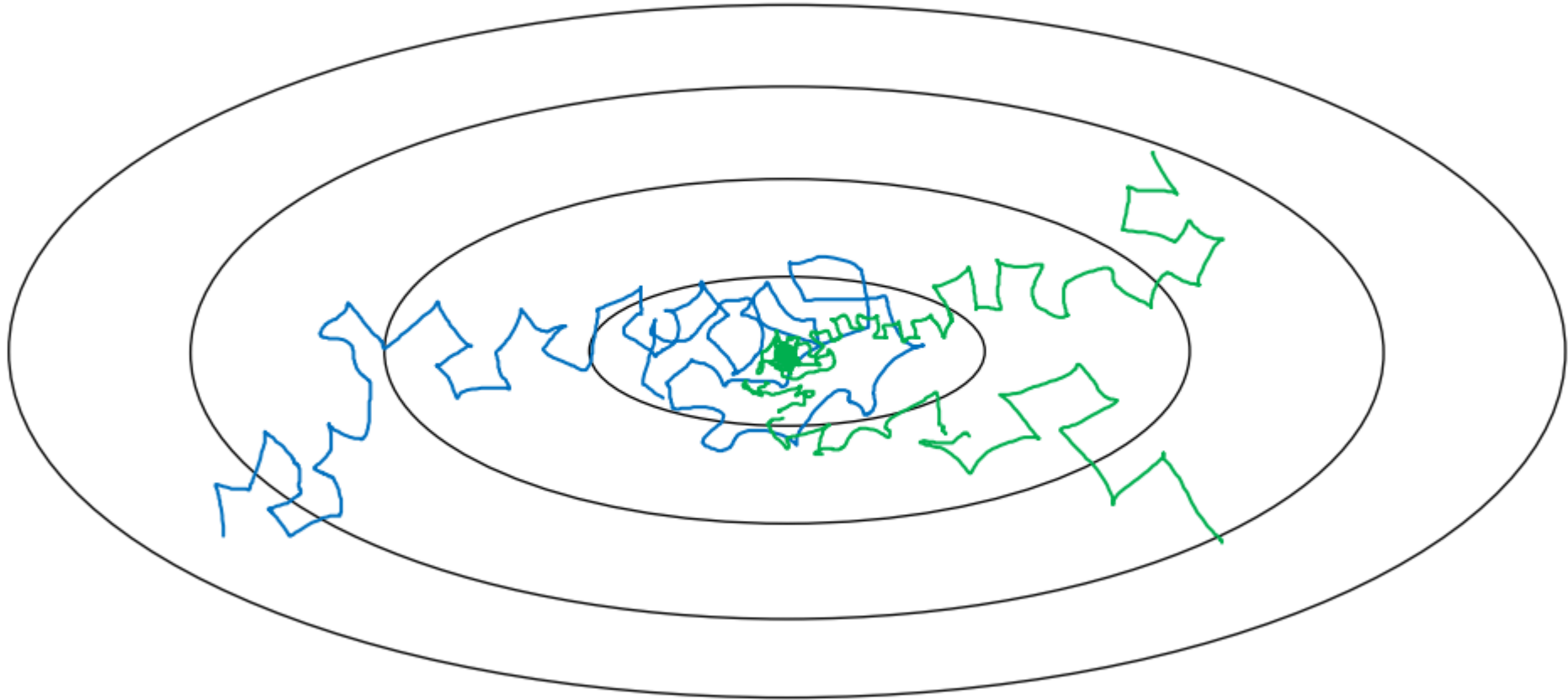
$$W = W - \alpha \frac{v_{dW}^{corrr}}{\sqrt{s_{dW}^{corrr} + \epsilon}}, b = b - \alpha \frac{v_{db}^{corrr}}{\sqrt{s_{db}^{corrr} + \epsilon}}$$

Hyper parameter choice

- Adam (Adaptive moment estimation)
- α needs to be tuned
- β_1 is by default 0.9
- β_2 is by default 0.999
- ε is by default 10^{-8}

Learning rate decay

- Slowly reduce learning rate: $\alpha = \alpha * 1 / (1 + \text{decay-rate} * \text{epoch-num})$



Other learning rate decay methods

- Exponential decay
- Discrete stair case
- Manual decay

Multi-class Classification

Softmax regression

Recognizing Multiple Classes



3

1

2

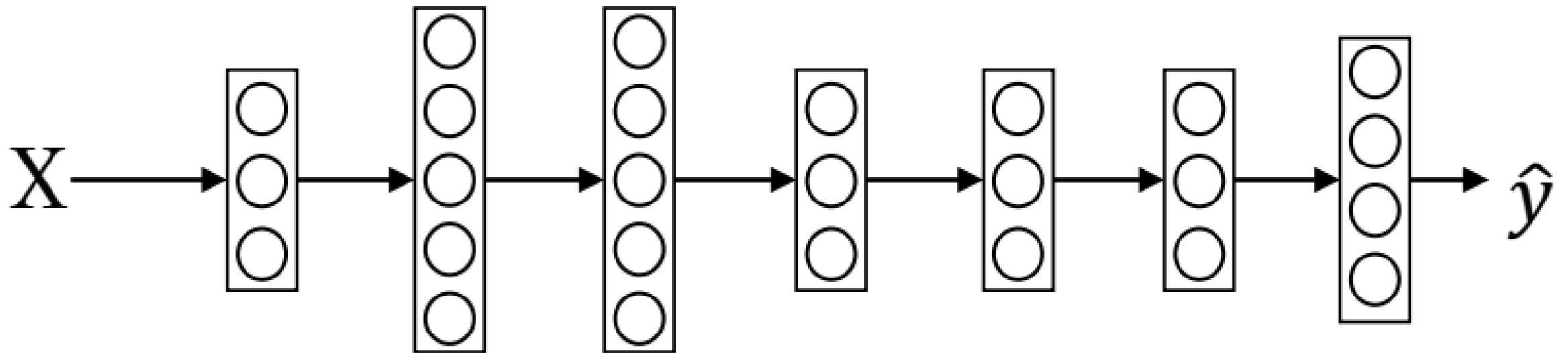
0

3

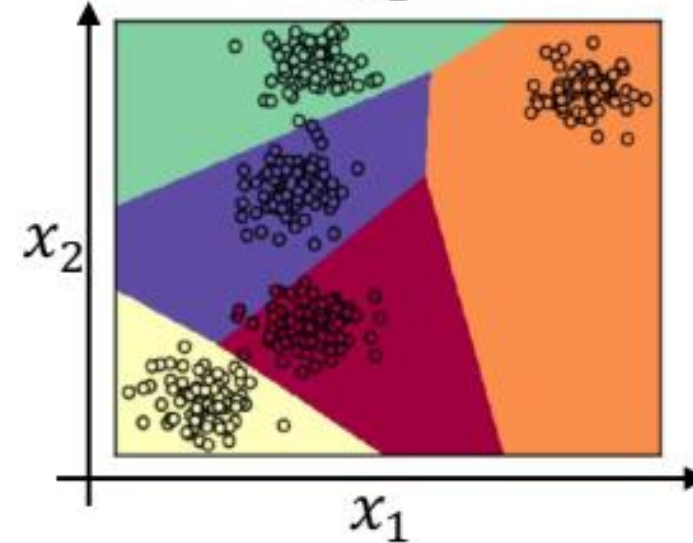
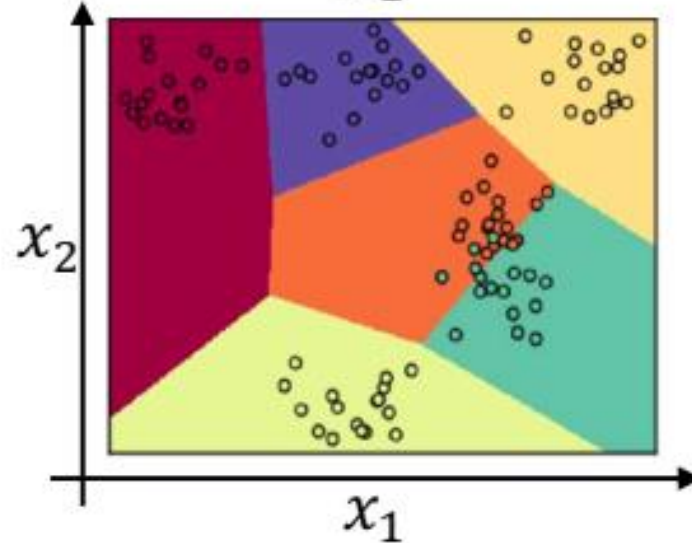
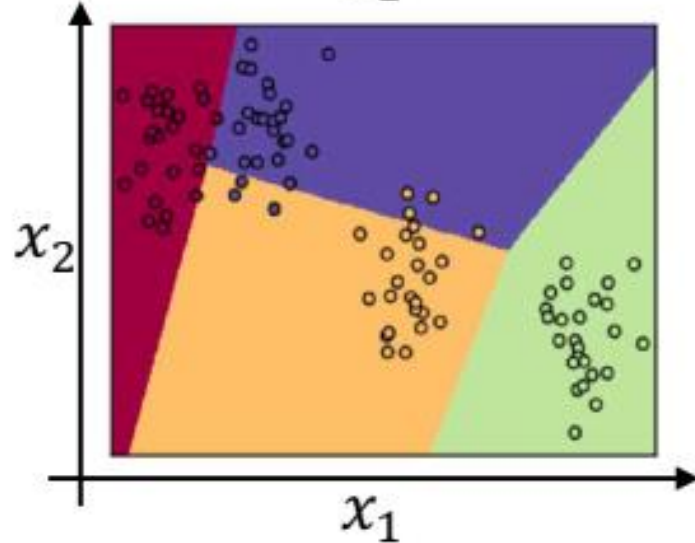
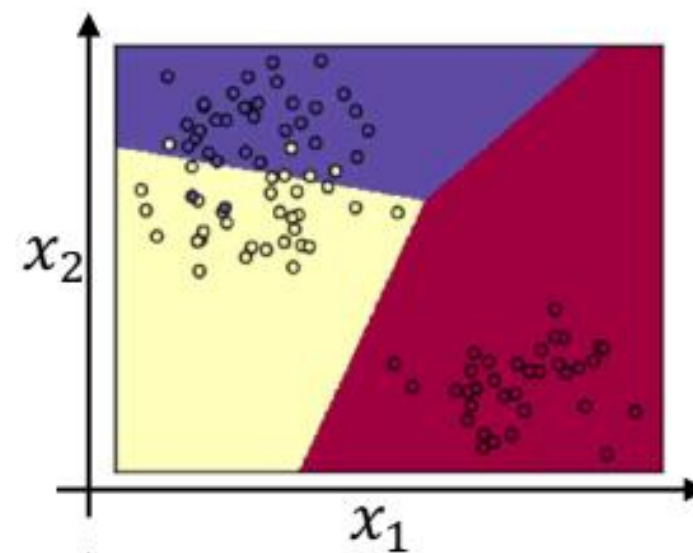
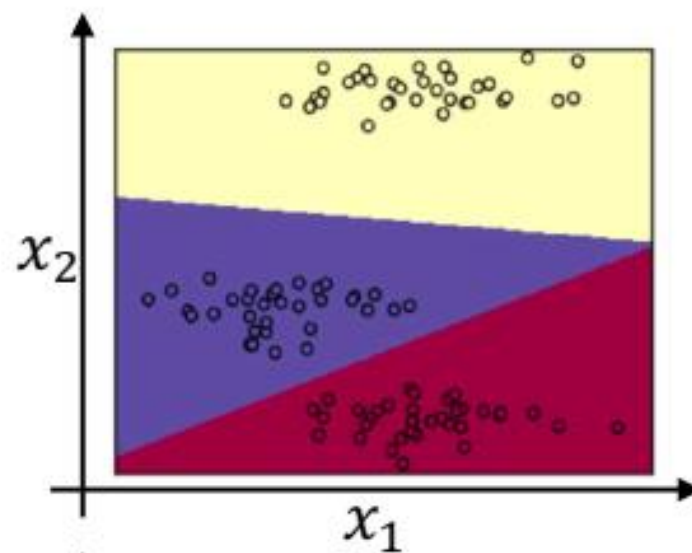
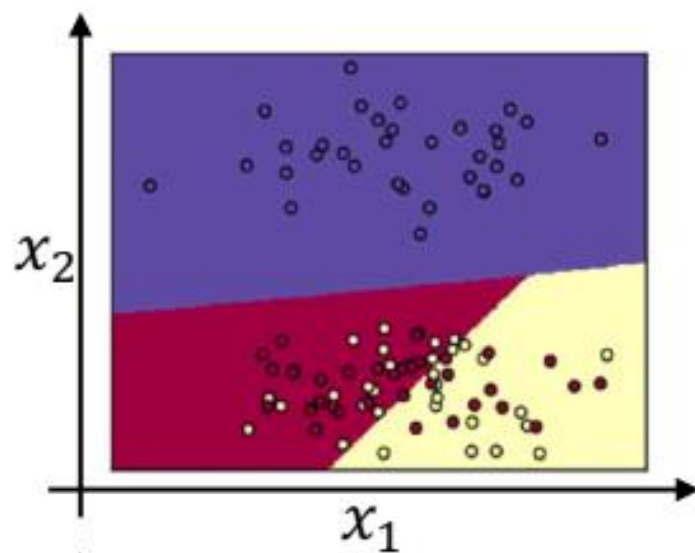
2

0

1



Softmax Examples



Loss Function

Pet	Cat	Dog	Turtle	Fish
Cat	1	0	0	0
Dog	0	1	0	0
Turtle	0	0	1	0
Fish	0	0	0	1
Cat	1	0	0	0

- $J = -\sum_{i=1}^k y_i \log(\hat{y}_i)$
- Assuming y is one hot encoding of the class label
- Use softmax as activation function
 - Transform output into probabilities

$$P(y=j \mid \theta^{(i)}) = \frac{e^{\theta^{(i)}}}{\sum_{j=0}^k e^{\theta_k^{(i)}}}$$

where $\Theta = w_0x_0 + w_1x_1 + \dots + w_kx_k = \sum_{i=0}^k w_ix_i = w^T x$

Softmax function



